INJECTIVE QUASI-COHERENT MODULES ON A NOETHERIAN SCHEME ARE FLASQUE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Sections of a quasi-coherent module over an open subset of a Noetherian scheme correspond to maps from a power of the quasi-coherent ideal sheaf of the closed complement. invoked at proof lines 1961-1965. Converts the extension problem for the section into an extension problem for maps.
- An injective object of the category of quasi-coherent modules admits extensions of maps along the inclusion of the ideal sheaf into the structure sheaf. used at proof lines 1965-1967. Produces the extending map from which the global section is read off.

Source reference. Original definition 01J4: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 1961-1963 to define the ideal sheaf I; the encoding of the section and the extension are inside the counted span.

Source reference. Original lemma 01YB: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 1963-1965 to encode s as sigma : I^n to J; the injectivity extension finishing the proof is inside the counted span.

Lemma 0.2. *Let X be a Noetherian scheme. Let $\mathcal{J}$ be an injective object of $QCoh(\mathcal{O}_X)$. Then $\mathcal{J}$ is a flasque sheaf of $\mathcal{O}_X$-modules.*

Proof. Let $U \subset X$ be an open subset and let $s \in \mathcal{J}(U)$ be a section. Let $\mathcal{I} \subset \mathcal{O}_X$ be the quasi-coherent sheaf of ideals defining the reduced induced scheme structure on $X \setminus U$ (see Schemes, Definition 01J4). By Cohomology of Schemes, Lemma 01YB the section s corresponds to a map $\sigma : \mathcal{I}^n \to \mathcal{J}$ for some n. As $\mathcal{J}$ is an injective object of $QCoh(\mathcal{O}_X)$ we can extend σ to a map $\tilde{s} : \mathcal{O}_X \to \mathcal{J}$. Then $\tilde{s}$ corresponds to a global section of $\mathcal{J}$ restricting to s. □

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